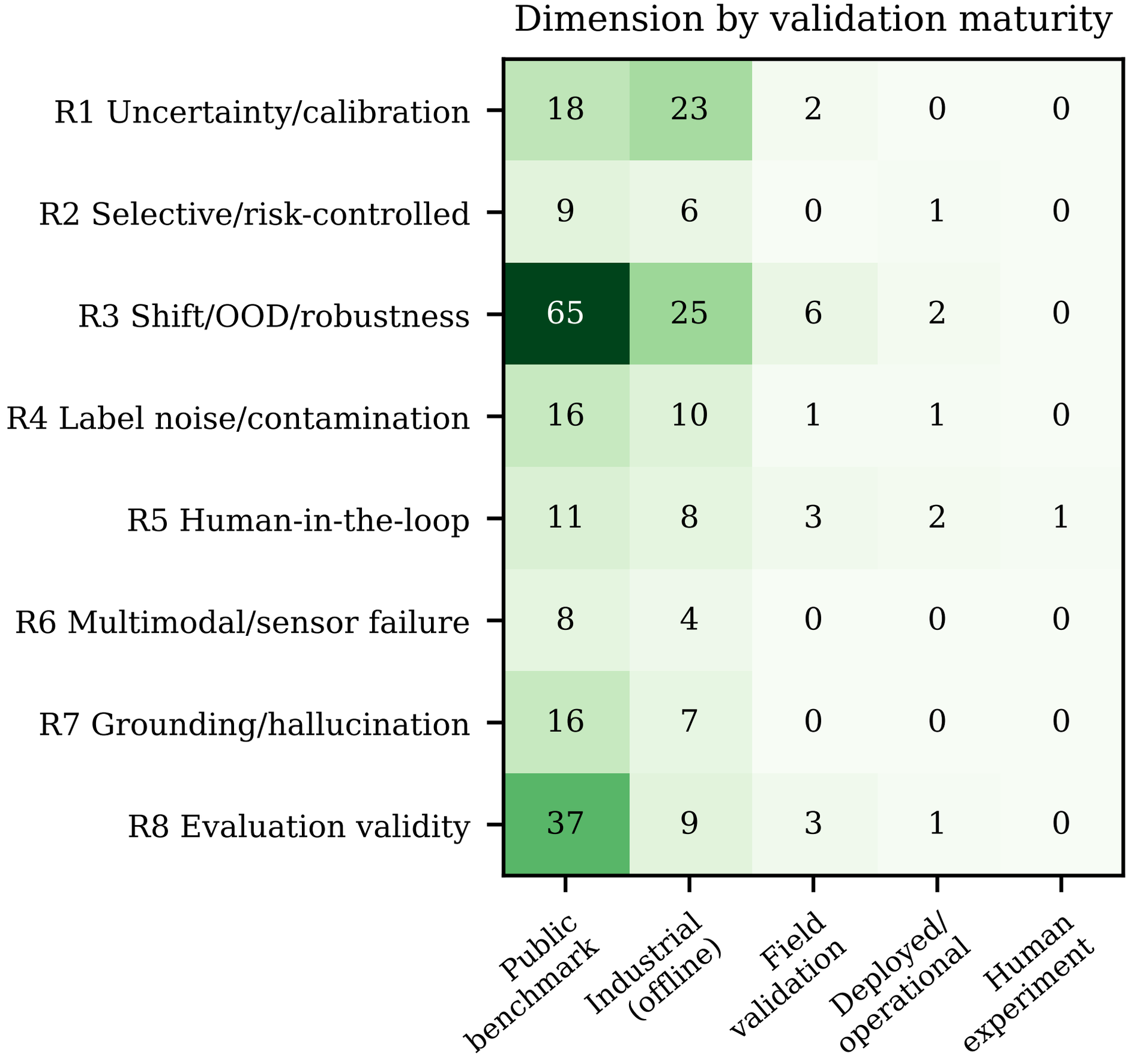
Dimension by validation maturity
R1 Uncertainty/calibration
R2 Selective/risk-controlled
R3 Shift/OOD/robustness
R4 Label noise/contamination
R5 Human-in-the-loop
R6 Multimodal/sensor failure
R7 Grounding/hallucination
R8 Evaluation validity
18 23 2 0 0
9 6 0 1 0
65 25 6 2 0
16 10 1 1 0
11 8 3 2 1
8 4 0 0 0
16 7 0 0 0
37 9 3 1 0
Public benchmark
Industrial (offline)
Field validation
Deployed/ operational
Human experiment